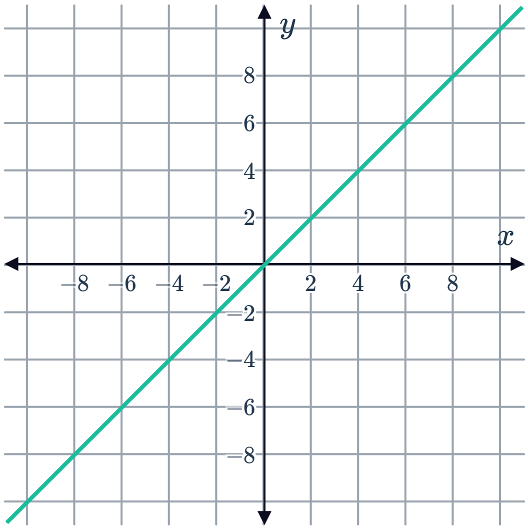
Gradient function graphs

1 Consider the graph of $y = x$:

a Find the gradient of the line at $x = 4$.

b Find the gradient at any value of x.

c What can be said about the gradient of a linear function?



2 Consider the graph of $y = x^2$ and its gradient function $y = 2x$:

a What can be said of the sign of the gradient function when $y = x^2$ is increasing or decreasing?

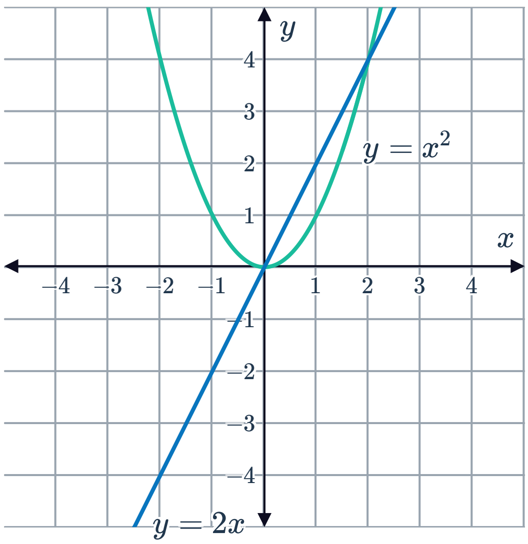
b For $x > 0$, is the gradient of the tangent positive or negative?

c For $x \geq 0$, as the value of x increases how does the gradient of the tangent line change?

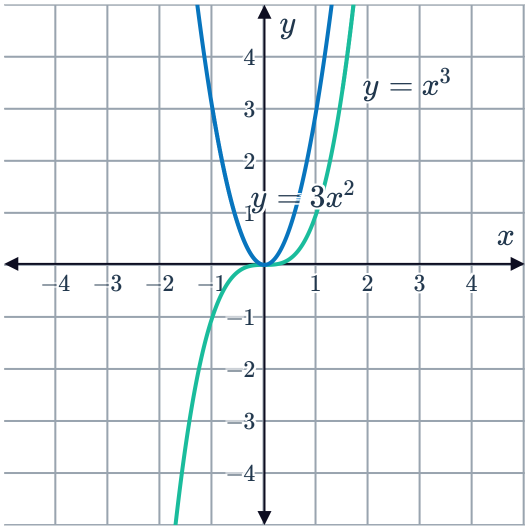
d For $x < 0$, is the gradient of the tangent positive or negative?

e For $x < 0$, as the value of x increases how does the gradient of the tangent line change?

f For $y = x^2$, the gradient of the tangent line changes at a constant rate. What type of function is the derivative of $y = x^2$?



3 Consider the graph of $y = x^3$ and its gradient function $y = 3x^2$.



a For $x > 0$, is the gradient of the tangent positive or negative?

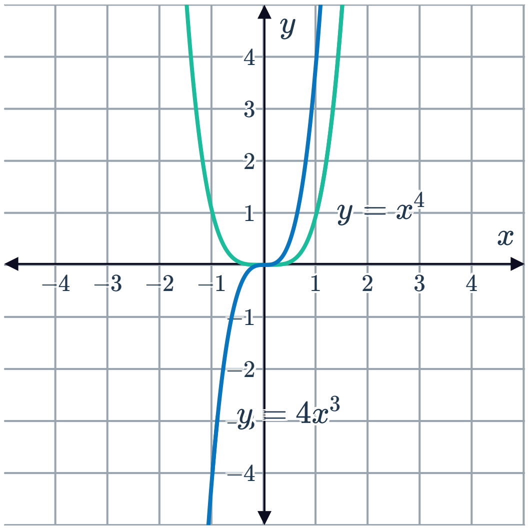
b For $x \geq 0$, as the value of x increases how does the gradient of the tangent line change?

c For $x < 0$, is the gradient of the tangent positive or negative?

d For $x < 0$, as the value of x increases how does the gradient of the tangent line change?

e For $y = x^3$, the gradient of the tangent line first decreases at a decreasing rate, then increases at an increasing rate. What type of function is the derivative of $y = x^3$?

4 Consider the graph of $y = x^4$ and its gradient function $y = 4x^3$.



a For $x > 0$, is the gradient of the tangent positive or negative?

b For $x \geq 0$, as the value of x increases how does the gradient of the tangent line change?

c For $x < 0$, is the gradient of the tangent positive or negative?

d For $x < 0$, as the value of x increases how does the gradient of the tangent line change?

e For $y = x^4$, the gradient of the tangent line is increasing, first at a decreasing rate and then at an increasing rate. What type of function is the derivative of $y = x^4$?

5 Consider the functions $f(x) = x^5$ and $g(x) = x^4$.

a Sketch the graph of $f(x)$ and its derivative.

b Sketch the graph of $g(x)$ and its derivative.

c If the degree of a function is even, will the degree of its derivative function be odd or even?

d If the degree of a function is odd, will the degree of its derivative function be odd or even?

Algebraic differentiation

6 Consider the function $f(x) = x^2$.

a Find the derivative of $f(x) = x^2$ from first principles.

b The derivative of $y = x^n$ for other values of n can be found using first principles in a similar way. The results for $n = 2, 3, 4$ and 5 are summarised in the following table:

y	x^2	x^3	x^4	x^5
y'	$2x$	$3x^2$	$4x^3$	$5x^4$

By observing the pattern in the table, find the derivative of the function $g(x) = x^8$.

c Write the general form of the derivative of $y = x^n$ for any value of n.

7

a For each of the following functions, state the degree of the gradient function:

i $y = x^2$ ii $y = x^3$ iii $y = x^4$

b Hence, state the degree of the derivative of a polynomial function of degree n.

8 Find the derivative of the following functions with respect to x:

a $y = x^7$ b $y = x^9$ c $y = x^{-7}$ d $y = x^{-1}$

e $y = x^{\frac{6}{5}}$ f $y = x^{\frac{2}{3}}$ g $y = -6$ h $y = x^{0.6}$

9 Consider the function $y = \sqrt{x}$.

a Rewrite the function in index form.

b Find the derivative of $y = \sqrt{x}$, in surd form.

10 Consider the function $y = \dfrac{1}{x^2}$.

a Rewrite the function in negative index form.

b Find the derivative of $y = \dfrac{1}{x^2}$.

11 Consider the function $y = \frac{1}{\sqrt[4]{x}}$.

a Rewrite the function in negative index form.

b Find the derivative of $y = \frac{1}{\sqrt[4]{x}}$, in positive index form.

12 Differentiate $y = t\sqrt{t}$.

13 Differentiate $y = x^{-\frac{1}{5}}$ with respect to x. Express your answer in positive index form.

Applications

14 Find the gradient of $f(x) = x^4$ at $x = 2$. Denote this gradient by $f'(2)$.

15 Consider the function $f(x) = x^9$.

a Find the gradient of the tangent to the function at $x = 1$.

b Find the equation of the tangent to the function at $x = 1$.

16 Consider the function $f(x) = \frac{1}{\sqrt{x}}$.

a Find the gradient of the tangent to the function at $x = 1$.

b Find the equation of the tangent to the function at $x = 1$.

17 David draws the graphs of x^2, x^3, x^4 and x^5 and draws the tangents to each one at the point where $x = 1$.

a David then notices where each of the tangents cut the y-axis and records this in the table below:

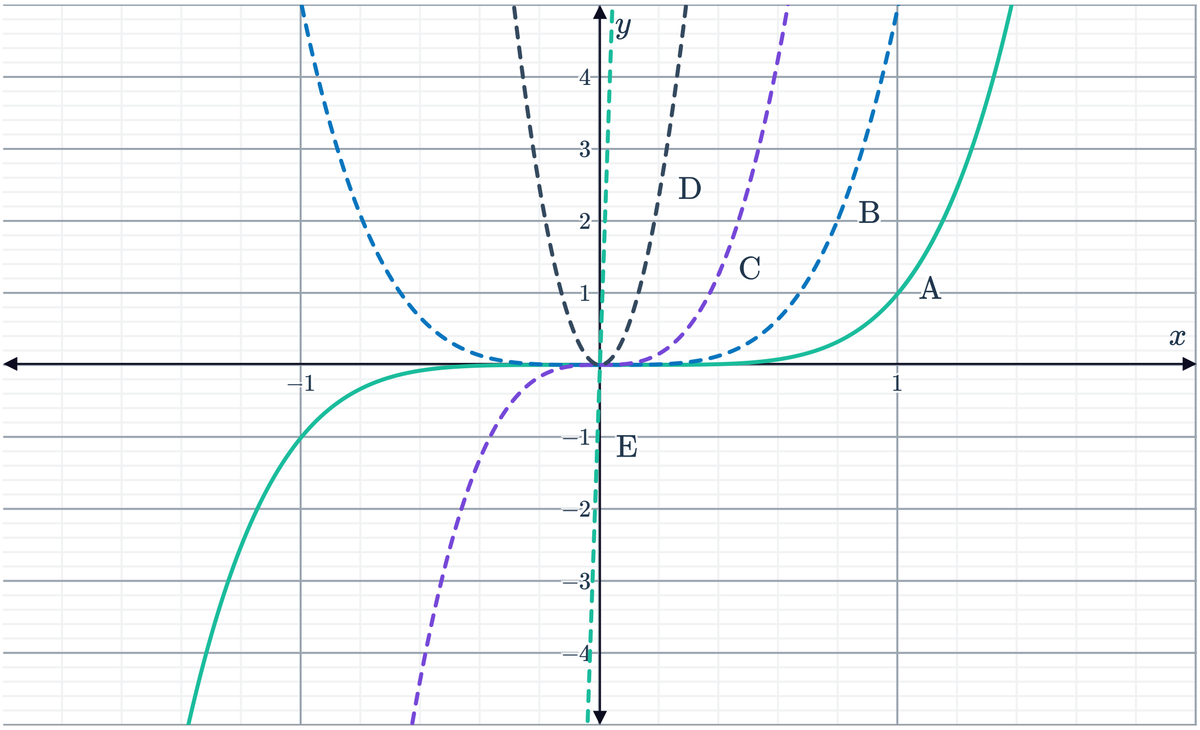
Graph	y-intercept of the tangent line	Gradient of the tangent
$y = x^2$	$(0, -1)$	
$y = x^3$	$(0, -2)$	
$y = x^4$	$(0, -3)$	
$y = x^5$	$(0, -4)$	

Complete the table by calculating the gradient of each of the tangents at $x = 1$.

b Following the pattern in the table, wha[redacted]ld be the gradient of the tangent to the graph of $y = x^n$ at the point where $x = 1$?

c Could the equation of the derivative of $y = x^6$ be $y' = x^5$? Explain your answer.

18 The graph of $y = x^5$ is shown below labelled as A. Fiona then graphs the derivative of the function, labelling it as B. She then finds the derivative of graph B to get graph C, then differentiates again to get graph D and differentiates again to get graph E.



State whether the following statements are true about this sequence of derivatives:

a The derivative of a function is always positive when the function is negative, and negative when the function is positive.

b Each graph is a function of the form ax^n.

c For any value of x, the value of the derivative will always be greater than the value of the function.

d The degree of the derivative is always different to the degree of the function.

19 Lucy used first principles to find the derivatives of $y = x^{\frac{1}{2}}$, $y = x^{\frac{3}{2}}$ and $y = x^{\frac{5}{2}}$. Her results are shown in the following table:

a Find the derivative of $y = x^{\frac{4}{3}}$.

b Find the derivative of $y = x^{\frac{3}{4}}$.

y	$x^{\frac{1}{2}}$	$x^{\frac{3}{2}}$	$x^{\frac{5}{2}}$
$\frac{dy}{dx}$	$\frac{1}{2}x^{-\frac{1}{2}}$	$\frac{3}{2}x^{\frac{3}{2}}$	$\frac{5}{2}x^{\frac{3}{2}}$